AutoApply Pro: Complete Technical Architecture Document

1. Technology Stack & Reasoning

Component	Choice	Reasoning
Backend	Java (Spring Boot 3)	Enterprise scalability and robust security.
Frontend	React (Vite + TS)	Fast performance and modular component ecosystem.
Database	PostgreSQL	Strong relational integrity for job and user tracking.
Storage	AWS S3	Scalable object storage for resumes.
Auth	JWT + Spring Security	Stateless authentication for decoupled architecture.
AI Service	Gemini API	High-performance content generation.

2. Feature-First Folder Structure

Backend (/backend/src/main/java/com/amrit/autoapply)

```
/common      - Global Exceptions, Utils
/config      - Security, WebMvc (CORS), Bean configs
/modules
  /auth      - AuthController, AuthService, SecurityConfig
  /jobs      - JobController, JobService, JobRepository
  /outreach  - OutreachController, EmailService (SMTP)
```

Frontend (/frontend/src)

```
/features
  /auth      - Login, AuthGuard, Hooks
  /dashboard - Stats, Views, DashboardService
  /jobs      - JobTable, ApplyButton, JobService
/components  - Shared UI (Buttons, Inputs, Modals)
/routes      - ProtectedRoute definitions
```

3. Database Schema Design (Fields)

- **users:** id, email, password_hash, role (Enum)
- **candidates:** id, user_id (FK), skills (JSON), job_preferences (JSON)
- **resumes:** id, candidate_id (FK), storage_url, is_primary, uploaded_at
- **companies:** id, name, industry
- **recruiters:** id, company_id (FK), name, email
- **jobs:** id, company_id (FK), title, description, status
- **outreach_history:** id, job_id (FK), template_used, sent_at, status

4. Configuration & Environment Variables

Use a `.env` file for the following:

- **Database:** DB_URL, DB_USER, DB_PASSWORD
- **Email:** SMTP_HOST, SMTP_PORT, SMTP_USER, SMTP_PASSWORD
- **Security:** JWT_SECRET, JWT_EXPIRATION
- **AI:** GEMINI_API_KEY
- **Storage:** AWS_S3_BUCKET_NAME, AWS_ACCESS_KEY, AWS_SECRET_KEY